

On-demand prediction of low-frequency average sound absorption coefficient of underwater coating using machine learning

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ABSTRACT

This study proposes an underwater coating with sound absorption ability in the middle-to-low frequency range and establishes an acoustic theoretical model combining the equivalent medium theory and the transfer matrix method. The sound absorption coefficient, surface characteristic impedance, equivalent volume longitudinal wave modulus, and equivalent sound velocity are calculated and solved. Using the preset 20 sensitive parameters and the hypercube sampling method, this study establishes 100,000 random sound absorption coefficient curves in the frequency range of 1 Hz–1,000 Hz. Further, deep neural networks are employed to predict the average value of the sound absorption coefficient curve. The overall loss function is derived by combining the mean square error between the expected average sound absorption coefficient and its predicted value and the network-optimized loss function to ensure that the 20 sensitive parameters that meet the acoustic performance can be predicted. Finally, two randomly selected sound absorption curves are used for prediction tests. The verification results indicate that the error between the expected average absorption coefficient and the predicted average absorption coefficient corresponding to the 20 sensitive parameters is only 0.026 % and 0.33 %. The proposed method can be extended to predict the average absorption coefficient value for any acoustic structure, which could be beneficial for the performance development of acoustic functional devices.

1. Introduction

One of the main development directions in the field of acoustic metamaterials [1–3] in recent years has been how to apply artificial intelligence to design acoustic structures and materials on demand [4], and machine learning-based methods have undoubtedly been verified as one of the effective means [5–7]. Currently, machine learning is the mainstream method for solving artificial intelligence-related problems, and as an independent direction in the field, it is developing rapidly. The key to combining machine learning with acoustic metamaterials lies in developing and applying artificial intelligence-based algorithms that can enable machines to learn from sample databases and improve the ability of metamaterials to control sound waves. The applications of machine learning to the fields of speech recognition [8], computer vision [9], natural language processing [10], material behavior [11], non-destructive evaluation [12] and healthcare [13] have achieved notable success in recent years. The design process of acoustic metamaterials involves many fields, including physics, acoustics, materials,

and engineering. In the past, theoretical calculations and finite element methods faced the limitation that a structure defines the performance [14]. However, this approach includes numerous trials, has high error and labor costs, and requires much time and computational resources, which is unfavorable for the rapid optimization design of acoustic metamaterials.

Regarding the design of acoustic materials, various machine learning-based methods have been proposed for optimizing acoustic structures and materials, and they can be mainly divided into the following categories. Supervised learning is a subfield of the machine learning field, and a support vector machine (SVM) is a powerful supervised learning-based model that has been widely used in classification and regression tasks. The SVM can be used in the design of acoustic metamaterials to classify the acoustic properties of materials and to predict the impact of different structural parameters on a material's performance, providing data-driven guidance for material design [15]. Unlike supervised learning, unsupervised learning does not use pre-labeled data to train models. On the contrary, it discovers hidden

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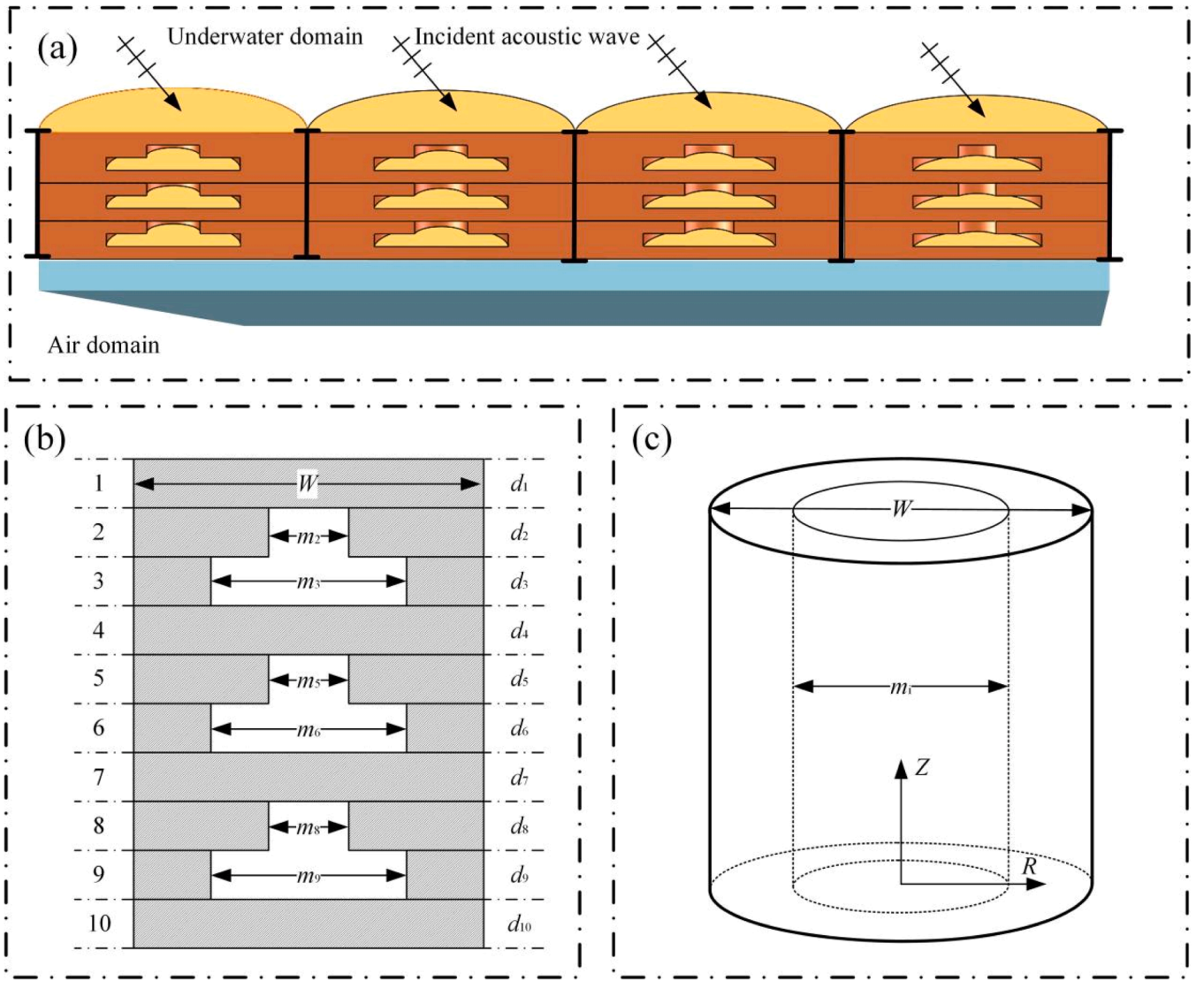


Fig. 1. The schematic diagram of geometric structural dimensions of the proposed underwater coating. (a) represents the arrangement and combination of underwater coating. (b) is the central section of the underwater coating. (c) represents the hollow cylindrical model of the second, third, fourth, fifth, eighth, and ninth layers in radial coordinates.

patterns in data by analyzing the features in a particular dataset. The principal component analysis (PCA) is a statistical technique that can reduce data dimensionality while preserving as much original data as possible. The PCA can be used to analyze and simplify complex datasets in acoustic metamaterial design optimization [16–18]. This helps identify the key factors affecting material properties, thus ensuring more effective material design and optimization. Reinforcement learning can be described as a combination of the optimization and decision-making processes, which lies between supervised and unsupervised learning. Reinforcement learning is highly suitable for multi-objective optimization and for solving high-dimensional parameter space problems in acoustic metamaterial design [19]. Design tasks typically involve finding an optimal solution in a complex parameter space by adjusting the periodic structures, geometric shapes, and other material parameters. For instance, reinforcement learning can effectively balance the objectives of acoustic transmission, isolation, and focusing performance [20–22]. The DL is a branch of machine learning [4,23–25] that can be applied to all three types of learning: unsupervised learning, supervised learning, and reinforcement learning. In deep reinforcement learning (DRL), deep neural networks (DNNs) can approximate value functions and optimize strategies. Compared to traditional machine learning

techniques, DL models can solve complex learning and decision-making problems by constructing DNNs [26–30]. This enables them to learn more complex feature representations and abstract concepts effectively. A convolutional neural network (CNN) is an artificial neural network [31–34] that has multiple advantages in processing grid-structured data. Unlike traditional fully connected neural networks, CNNs integrate spatial structural information on input data by introducing convolutional layers, pooling layers, and local connections.

Recently, the design of acoustic metamaterials has experienced a shift toward more complex model architectures, such as neural networks, CNNs, DRLs, and GANs. These DL-based models for optimizing acoustic metamaterial structures show significant advantages and applicability in practical tasks. The efficiency and performance of acoustic metamaterial design can be improved by defining appropriate method requirements and task objectives based on specific design demands. For instance, a CNN-based prediction method for an ultra-thin metasurface acoustic absorber, which can significantly reduce the characterization time while maintaining accuracy, was developed in [35]. Based on deep learning, broadband acoustic metamaterials operating within the target frequency band were designed, and it was confirmed that an average absorption coefficient was higher than 97 %

in the ultra-wideband range of 860 Hz to 8000 Hz [7]. In addition, a DNN-based inverse prediction mechanism was used to design low-frequency absorption absorbers based on Helmholtz resonators [36], and the trained convolutional neural network was used to predict the optimal design parameters based on the desired absorption characteristics with high fidelity in reverse. An improved Gaussian-Bayesian model was effectively applied to the design of machine learning-assisted subwavelength absorbers with coherent coupling weak resonance, ensuring flexibility, efficiency, and broadband characteristics at frequencies lower than 500 Hz [37]. Further, a machine learning-based deep autoencoder (DAE) model was used to predict absorption peak points and highly complex acoustic structures [30].

Different from the above airborne acoustic metamaterials, the design of underwater acoustic metamaterials, especially for the design of underwater coating, improving sound absorption performance is a challenging target due to underwater impedance mismatch and underwater pressure resistance limitations. In addition, due to the high production and experimental costs of underwater coating, accurate estimation of their acoustic properties is required before preparing experimental samples, especially predicting the structural and material parameters corresponding to the expected average sound absorption coefficient within a specific frequency band, which is a key factor. Previous studies employed the method of introducing prior knowledge into data-driven structural optimization at low frequencies and reduced the average acoustic transmission loss by 31.78 % compared to traditional impedance gradient design [38]. In addition, machine learning-based and optimization algorithms were combined with finite element models to realize an accelerated reverse design of acoustic coatings in underwater sound attenuation applications; this approach increased the prediction speed of absorption coefficients by 4500 times and provided high-performance suboptimal solutions rapidly and efficiently. In earlier studies, artificial intelligence methods were not used to predict the average value of sound absorption performance. However, for the sound absorption performance of acoustic metamaterials, the average sound absorption coefficient in a certain frequency band represents the strength of their ability to absorb incident sound waves. This is of great help for the rapid and intelligent design of acoustic metamaterials.

Starting from predicting the structural dimensions and material properties of the underwater acoustic cover layer in the mid to low frequency range, especially within 1000 Hz, firstly, this study proposes a theoretical calculation method for an underwater coating structure with embedded multi-layer cavities and uses 100,000 sound absorption curves as a sample database. Secondly, 20 sensitive parameters are predicted based on the average value of a randomly selected sound absorption coefficient curve in the target frequency range of 1 Hz–1000 Hz. Thirdly, it is shown that the average sound absorption coefficient corresponding to the predicted sensitive parameters has a very small error compared to the average value of the randomly selected sound absorption coefficient curve. Finally, the proposed machine learning-based method could accelerate the development process of underwater acoustic materials and provide a convenient means for underwater acoustic manipulation. This study is organized as follows. In section2, the structure and materials of underwater coating, as well as the calculation methods for acoustic properties, are presented in detail. Moreover, Section 2 gives out the preselected 100,000 representative sound absorption curves from the sample database. Section 3 presents deep learning-based framework and corresponding method. Section 4 provides the predicted results and errors for the average of any two absorption curves. Finally, we present conclusions in Section 5. The contribution of this study lies in its valuable reference for the rapid estimation and on-demand design of acoustic performance of underwater coatings, which improves the development efficiency of acoustic performance of underwater functional devices. Machine learning methods to predict the inverse structural parameters of a given average sound absorption coefficient also provides assistance for the development of more acoustic materials. It is a manifestation of the application

Table 1

Specific structural sizes of units expressed in mm.

W	d_1	d_2	d_3	d_4	d_5	d_6	d_7	d_8
2E3	10	10	10	10	10	10	10	10
d_9	d_{10}	m_2	m_3	m_5	m_6	m_8	m_9	
10	10	1E3	1E3	1E3	1E3	1E3	1E3	

Table 2

Mechanical parameters of the units.

Density ρ	Young's modulus E	loss factor η	Poisson's ratio ν
1130 kg/m ³	5E7(1+ η) Pa	0.4	0.44

of artificial intelligence methods in acoustic materials or structures.

2. Acoustic theoretical models and databases

This study proposes an underwater coating that can achieve good sound absorption performance in the low-to-middle frequency range. The periodic array combination is shown in Fig. 1a, and the proposed underwater coating with rigid boundaries between the units is displayed in Fig. 1. A rigid backing is placed at the bottom of the underwater coating to achieve complete reflection conditions. The incident sound waves in the water enter the underwater coating units from top to bottom. Each unit is divided into 10 layers, where the second, third, fifth, sixth, eighth, and ninth layers are hollow structures. The geometric structure parameters of the underwater coating are shown in Table 1; the units' substrate material is rubber, and their material parameters are presented in Table 2.

As shown in Figs. 1a and 1b, there are six layers of hollow structures inside the underwater coating, which can be considered uniform rubber layers with equivalent parameters. Accordingly, the structure presented in Fig. 1b can be regarded as a combination of multiple layers of rubber with different equivalent parameters. In this study, each layer of a hollow structure is considered a mechanical model of a hollow circular tube, as displayed in Fig. 1c. Therefore, an elastic wave excited by the incident sound wave entering the structure of underwater coating has the same direction as the incident sound wave, and the dynamic elastic modulus in rubber can be expressed using the Lamé constant λ and complex shear modulus μ , which are respectively defined as follows:

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \quad (1)$$

$$\mu = \frac{E}{2(1+\nu)}, \quad (2)$$

In the cylindrical coordinate system, the rubber layer of the hollow structure can be treated as a pipeline with viscoelastic properties, and the corresponding mechanical equations are as follows:

$$\rho \frac{\partial^2 u_r}{\partial t^2} = (\lambda + 2\mu) \frac{\partial}{\partial r} \left[\frac{1}{r} \frac{\partial (ru_r)}{\partial r} \right] + (\lambda + \mu) \frac{\partial^2 u_r}{\partial r \partial z} + \mu \frac{\partial^2 u_r}{\partial z^2} \quad (3)$$

$$\rho \frac{\partial^2 u_z}{\partial t^2} = (\lambda + 2\mu) \frac{\partial^2 u_z}{\partial z^2} + \frac{(\lambda + \mu)}{r} \frac{\partial^2 (ru_z)}{\partial r \partial z} + \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_z}{\partial r} \right) \quad (4)$$

where u_r and u_z are the vibration displacements along the R and Z directions, respectively.

At the hollow part's interface, that is, the radius boundary of $r = m_i/2$ ($i = 2, 3, 5, 6, 8, 9$) is a free surface, and normal stress σ_{rz} and shear stress τ_{rz} both equal zero. Due to the fixed installation of underwater coating in practical use, the outer boundary $r = w/2$ is stationary, and u_r and shear stress τ_{rz} are both zeros. Further, normal stress σ_{rz} and shear stress τ_{rz} in cylindrical coordinates can be respectively expressed by:

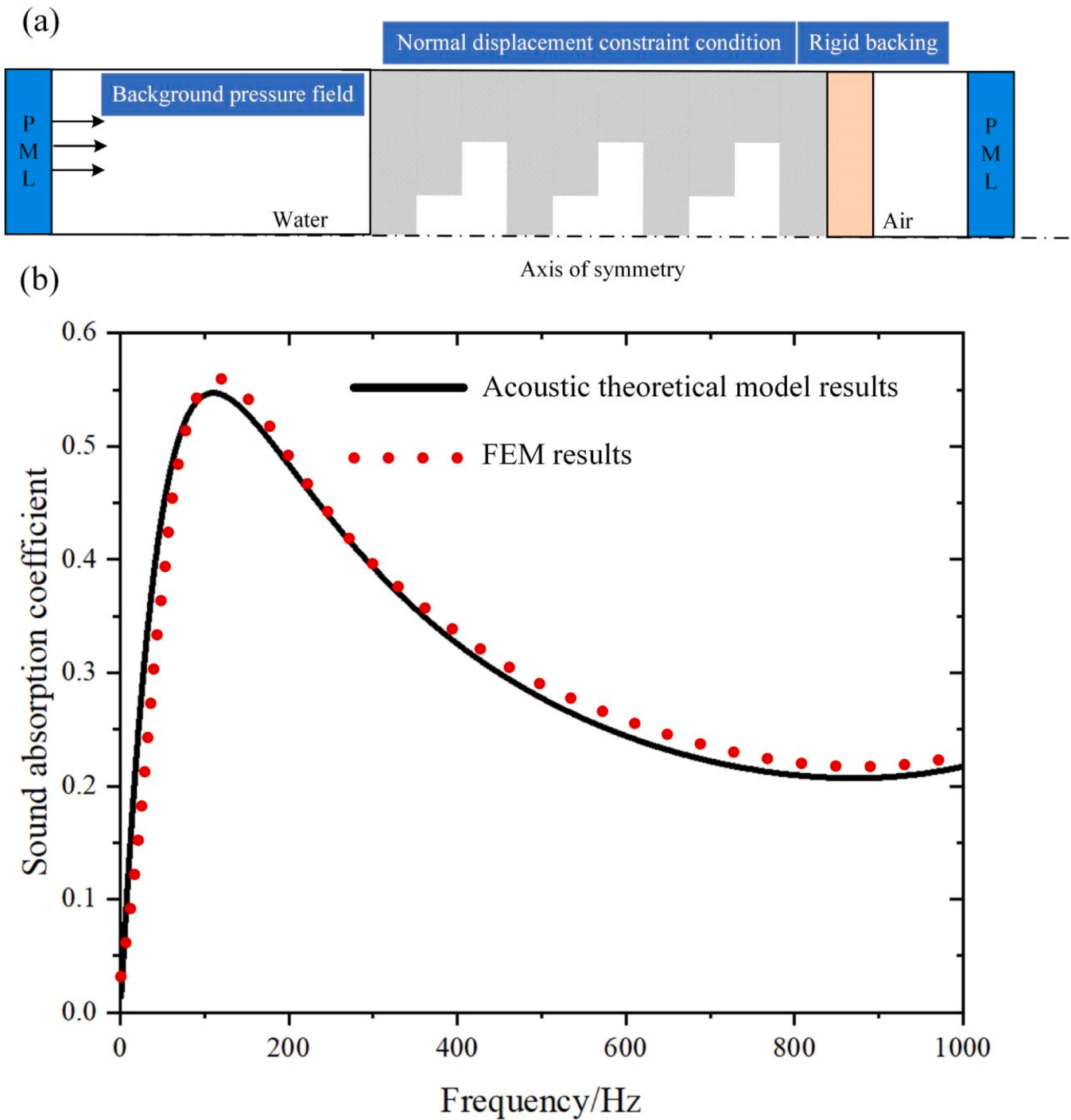


Fig. 2. The sketch map of the FEM method (a) and the sound absorption coefficient calculation results (b), red dotted line and black solid line represent FEM and acoustic theoretical calculations, respectively.

$$\sigma_{rz} = \lambda \left[\frac{1}{r} \frac{\partial(ru_r)}{\partial r} + \frac{\partial u_z}{\partial z} \right] + 2\mu \frac{\partial u_z}{\partial r}, \quad (5)$$

$$\tau_{rz} = \mu \left[\frac{\partial u_z}{\partial r} + \frac{\partial u_r}{\partial z} \right], \quad (6)$$

In this study, the harmonic plane waves propagate through the underwater coating along the Z-direction, which can be expressed by:

$$u_z = Ae^{j(\omega t - k'z)}, \quad (7)$$

where k' is the equivalent complex wave number, which includes

longitudinal wave number k_l and shear wave number k_τ , and it is defined by:

$$k_l = \sqrt{\frac{\rho\omega^2}{\lambda + 2\mu}}, k_\tau = \sqrt{\frac{\rho\omega^2}{\mu}}, \quad (8)$$

By substituting Eq. (7) into Eq. (4), u_r and u_z can be solved, and an equivalent complex wave number k' can be obtained. By combining Eqs. (9) and (10) with acoustic boundary conditions, a homogeneous system of equations with undetermined coefficients C1 – C4 can be obtained, and further the characteristic equation containing k' can be obtained

$$u_z = C_1 J_0 \left(r \sqrt{k_i^2 - k^2} \right) + C_2 N_0 \left(r \sqrt{k_i^2 - k^2} \right) + C_3 J_0 \left(r \sqrt{k_i^2 - k^2} \right) + C_4 N_0 \left(r \sqrt{k_i^2 - k^2} \right) \quad (9)$$

$$u_r = -jC_1 \frac{\sqrt{k_i^2 - k^2}}{k'} J_1 \left(r \sqrt{k_i^2 - k^2} \right) - jC_2 \frac{\sqrt{k_i^2 - k^2}}{k'} N_1 \left(r \sqrt{k_i^2 - k^2} \right) + jC_3 \frac{k'}{\sqrt{k_i^2 - k^2}} J_1 \left(r \sqrt{k_i^2 - k^2} \right) + jC_4 \frac{k'}{\sqrt{k_i^2 - k^2}} N_1 \left(r \sqrt{k_i^2 - k^2} \right) \quad (10)$$

This study defines $\varepsilon = m_i/w$ as the perforation rate of a layer, and the volume longitudinal wave modulus S' , the equivalent density ρ' , and the equivalent complex sound velocity c' are respectively expressed by:

$$S' = \frac{\mu(\lambda + 2\mu)(\varepsilon^2 + 1) + 2\varepsilon^2 \lambda}{(\lambda + \mu)\varepsilon^2 + \mu}, \quad (9a)$$

$$\rho' = \rho(1 - \varepsilon^2), \quad (10a)$$

$$c' = \sqrt{\frac{S'}{\rho'}}, \quad (11)$$

After obtaining the equivalent parameters of each layer of underwater coating, the transfer matrix method is used to solve the sound absorption coefficient. Then, the transfer matrix $[T_i]$ of the i th layer is expressed by:

$$[T_i] = \begin{bmatrix} \cos(k_i d_i) & j\rho' c' \sin(k_i d_i) \\ \frac{j \sin(k_i d_i)}{\rho' c'} & \cos(k_i d_i) \end{bmatrix}, i = 1, 2, \dots, 10, \quad (12)$$

Considering the requirement for equivalent sound pressure continuity and equivalent volume velocity continuity between the layers, as well as the condition of incident wave perpendicular incidence, the total matrix $[T]$ can be expressed by:

$$[T] = [T_1][T_2][T_3] \dots [T_{10}], i = 1, 2, \dots, 10, \quad (13)$$

The bottom end of the underwater coating studied in this work is set as a rigid backing. Then, the surface impedance Z_{in} of the overall structure can be calculated using Eqs. (12) and (13), which can be further used to determine the complex reflection coefficient of the underwater coating as follows:

$$R = \frac{Z_{in} - \rho_w c_w}{Z_{in} + \rho_w c_w}, \quad (14)$$

To verify the correctness of the above acoustic theoretical calculations, we used finite element method (FEM) to calculate the sound absorption coefficient of the underwater coating in Fig. 1. In order to improve computational efficiency, the FEM simulation calculation of underwater coating is carried out using the two-dimensional axisymmetric model in COMSOL Multiphysics version 6.2 [39,40,41]. The FEM model is shown in Fig. 2(a), where the incident sound wave is applied to the left water area in the form of a background pressure field. The dotted line below is the axis of symmetry of the model, and normal displacement constraints are applied to the side boundaries. In the entire model, the water and air domains are fluid domains, and the pressure acoustic module is used for modeling. The acoustic cover layer and rigid backing are solid media, and the solid mechanics module is used for modeling. Apply acoustic structural interaction boundary conditions at the boundary between solid and fluid regions. Add a Perfect Match Layer (PML) to the outer boundary of the water body. The principle of solving using finite element method is as follows.

In a uniform incompressible ideal fluid medium, the acoustic wave equation is:

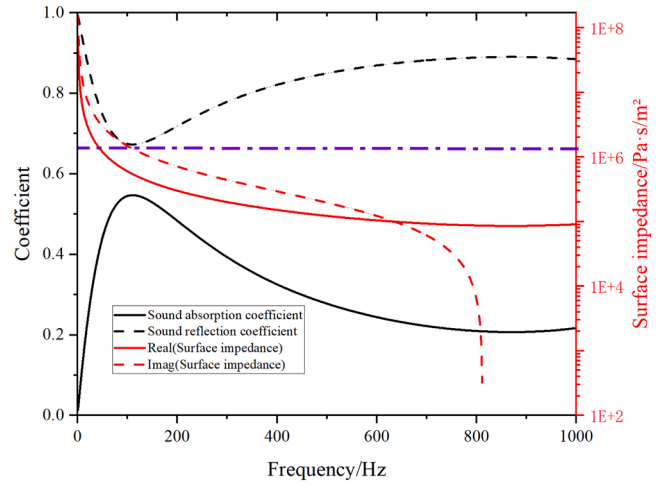


Fig. 3. The imaginary and real parts of the sound absorption coefficient, reflection coefficient, and surface impedance of the underwater coating.

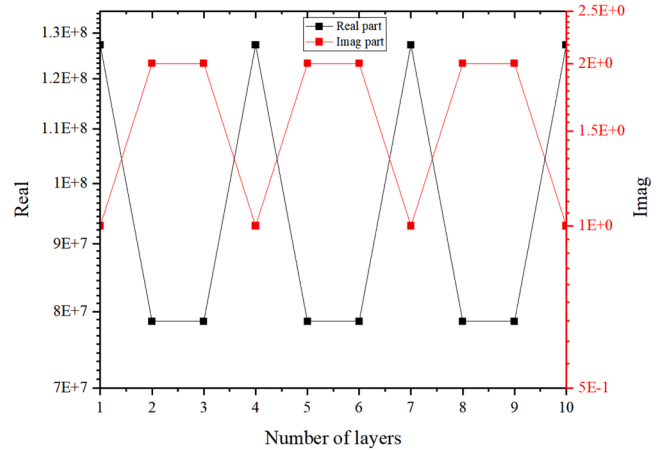


Fig. 4. The real and imaginary parts of the equivalent longitudinal wave modulus of each layer of the underwater coating.

$$\frac{1}{c} \frac{\partial^2 p}{\partial t^2} - \nabla^2 p = 0 \quad (15)$$

Where c is the speed of sound in the fluid medium, p is the sound pressure, and t is time. By solving the motion equation and acoustic/structural coupling equation in the fluid domain [38], the reflection coefficient R and transmission coefficient T can be obtained. Therefore, only (0,0) order waves are considered in this study, and the absorption coefficient α is calculated as follows:

$$\alpha = 1 - R^2 - T^2 \quad (16)$$

From Fig. 2, it can be seen that the FEM calculation results are almost consistent with the acoustic theoretical calculations in the frequency range of 1 Hz–1000 Hz, further confirming the correctness of the acoustic theoretical calculations. The absorption coefficient curve is presented in Fig. 3, where it can be seen that in the range of 1 Hz–1000 Hz, the absorption coefficient has a peak around 100 Hz, and the absorption coefficient throughout the entire frequency band is <0.6 . The reflection and absorption coefficients have symmetrical but opposite variation patterns. As shown in Fig. 3, the real and imaginary parts of the surface impedance of the overall structure gradually decrease with the frequency. The purple dashed line in Fig. 3 represents the characteristic impedance of the sound waves in aqueous media. The results indicated that when the real and imaginary parts of the surface impedance of the

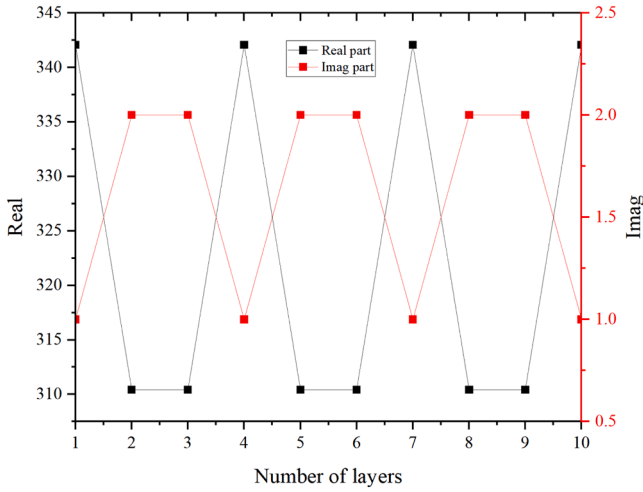


Fig. 5. The real and imaginary parts of the equivalent sound velocity of each layer of the underwater coating.

structure were close to the purple dashed line, the sound energy could relatively easily enter the underwater coating, and the corresponding sound absorption coefficient was the highest, whereas the reflection coefficient was the lowest.

The real and imaginary parts of the equivalent longitudinal wave bulk modulus and equivalent sound velocity were calculated for each layer of the underwater coating, as shown in Figs. 4 and 5. Due to the presence of a hollow portion, the real parts of the equivalent bulk modulus and equivalent sound velocity decreased in the hollow portion, whereas the imaginary parts increased. Next, the absolute value of the equivalent sound velocity was calculated for each layer of the underwater coating. The calculation results are shown in Table 3, where it can be seen that the absolute value of the equivalent sound velocity was the smallest in the hollow part. It can be further seen from Fig. 5 that both the real and imaginary parts of the equivalent sound velocity are smaller than the sound velocity in water. The equivalent sound velocity depends on Eqs. (9)–(11), mainly because underwater coating is a loss medium for underwater sound waves, however in underwater coating, the increase in perforation rate reduces the equivalent sound velocity. Therefore, for coatings with viscoelastic damping mechanical properties of the substrate material, the decrease in equivalent sound velocity means that the propagation distance and edge length of the sound wave in the coating will increase the loss, which is also the physical essence of underwater sound absorption.

Further, a sample database containing 16 sensitive geometric parameters and four sensitive material parameters directly affecting the sound absorption performance was constructed. These sensitive parameters included: (1) the thickness of each of the 10 layers in the underwater coating d_i , which was $1 \text{ mm} \leq d_i \leq 20 \text{ mm}$, where $1 \leq i \leq 10$; (2) the diameter of the hollow part m_i , which satisfied the condition of $20 \text{ mm} \leq m_i \leq 1980 \text{ mm}$, where $i = 2, 3, 5, 6, 8, 9$; (3) density ρ , which was $1\text{E}3 \text{ kg/m}^3 \leq \rho \leq 1.5\text{E}3 \text{ kg/m}^3$; (4) loss η , which satisfied the condition of $0.1 \leq \eta \leq 0.8$; (5) Young's modulus E , which was $1\text{E}7(1+\eta) \text{ Pa} \leq E \leq 1\text{E}8(1+\eta) \text{ Pa}$; (6) Poisson's ratio ν , which satisfied the condition of $0.4 \leq \nu \leq 0.49$. A database sample within the frequency range of 1 Hz–1000 Hz was established based on variables corresponding sensitive parameters. The more evenly distributed the data in the database were, the larger the range of data and the higher the accuracy of prediction

were. For frequencies ranging from 1 Hz to 1000 Hz, 20 sensitive parameters in each group corresponded to a sound absorption curve. In addition, 100,000 different sound absorption curves were constructed, and the selection method for data establishment adopted the Latin hypercube sampling (LHS) method, which is a commonly used random stratified sampling method in multivariate parameter distributions. The distribution of 100,000 sets of row data is displayed in Fig. 6. Since it was challenging to achieve a sound absorption coefficient above 0.8 at frequencies of below 400 Hz, this study used the dataset in Fig. 6 for deep learning training.

3. Deep learning-based framework and method

The average value of 100,000 absorption coefficient curves was calculated, and 20 sensitive parameters were predicted based on the randomly selected average absorption coefficient. The frequency was specified to be in the range of 1 Hz–1000 Hz. The method of training the sample database adopted the DNN introduced in [4], which can be regarded as a mapping function between the input data and corresponding output data. In this study, DNNs were used to estimate the corresponding sensitive parameters that could meet performance requirements from a selected average sound absorption coefficient. Typically, before data are fed to a deep learning model's input layer, it is necessary to normalize both the input and output data to accelerate the convergence speed of the model and improve network stability. After feeding the normalized input data to the model's input, the network provided the predicted normalized output; then, the inverse normalization was performed to obtain the non-normalized estimated value of the output data. It should be noted that the network input was the average sound absorption coefficient, whose value was from zero to one, so normalization was not necessary for these data; therefore, only the network output data, namely the structural parameters, were normalized using the min-max normalization method, which maps data of different dimensions to the numerical range of [0,1].

Assume that the model's normalized input data were denoted by $\bar{\alpha}$, the corresponding normalized true sensitive parameter data were denoted by s_n , and the mapped neural network model was defined by $f(\cdot)$; then, the data mapping process could be simply expressed as $s_n = f(\bar{\alpha})$. For such types of mapping tasks, feed-forward, fully connected neural networks, which are also known as multi-layer perceptrons or directly referred to as DNNs, are commonly used. The DNNs are typically

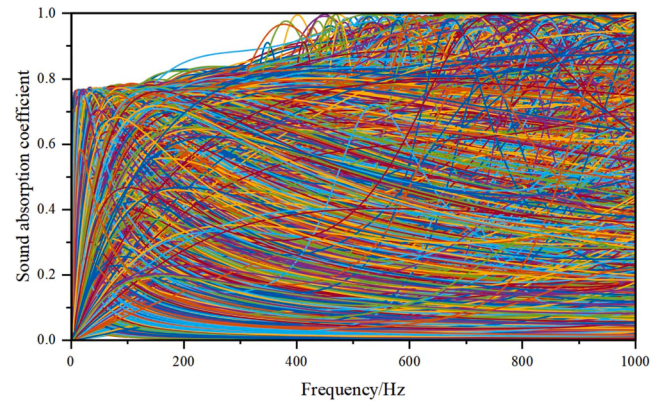


Fig. 6. The preselected 100,000 representative sound absorption curves from the sample database.

Table 3

The absolute value of the equivalent sound velocity of each layer of the underwater coating expressed in m s^{-1} .

Layer 1	Layer 2	Layer 3	Layer 4	Layer 5	Layer 6	Layer 7	Layer 8	Layer 9	Layer 10
348.34	316.12	316.12	348.34	316.12	316.12	348.34	316.12	316.12	348.34

Table 4

The parameters of the DNN model used in this study.

Network input	Normalized input data (average sound absorption coefficient $\bar{\alpha}$)
Network structure	128 fully connected layer - LeakyReLU
	128 fully connected layer - LeakyReLU
	128 fully connected layer - LeakyReLU
	128 fully connected layer - LeakyReLU
	20 fully connected layer - Sigmoid
Network output	Normalized sensitive parameters (20 in total)

multi-layered structures, where each layer consists of a certain number of interconnected neurons, and each neuron is essentially composed of a linear transformation and a nonlinear activation function. In the DNN structure, there are generally three types of layers: input, output, and hidden layers; the hidden layers are in the middle of the network between the input and output layers. The number of hidden layers and the number of nodes in them defined the modeling ability and complexity of a DNN network. For the j th hidden layer h^j , whose input is denoted by x , the input-output data mapping process can be expressed by:

$$h^j(x) = \sigma(w^j x + b^j), \quad (15a)$$

where w^j and b^j are the weights and biases of the hidden layer, respectively; σ is a nonlinear activation function.

The network structure used in this study is presented in Table 4. The entire network consisted of five fully connected layers, where the first four layers contained 128 neurons and had the LeakyReLU activation function defined by:

$$\sigma(x) = \begin{cases} x, & x \geq 0, \\ \theta x, & x < 0 \end{cases} \quad (16a)$$

Usually, parameter θ has a value of 0.01. The last hidden layer consisted of 20 neurons, and the normalized output values of the sensitive parameters corresponded to their respective outputs. The activation function of this layer was Sigmoid, which was defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \quad (17)$$

The loss function of network optimization was the mean square error (MSE), and it was calculated using the sensitive parameters $\hat{s}_n$ predicted by the network and their corresponding true values s_n as follows:

$$L_s = \frac{1}{M} \sum_{i=1}^M (s_{ni} - \hat{s}_{ni})^2, \quad (18)$$

Since materials with different sensitive parameters might correspond to the same average sound absorption coefficient, inputting a set of performance parameters in the reverse design might correspond to multiple sensitive parameters, which could prevent DNNs from

obtaining optimal results. Therefore, this study introduced additional constraints to the DNN model. In addition, from the perspective of the loss function, it was expected the network would provide sensitive parameters that could comply with the performance parameters, and only the MSE loss of the sensitive parameters was used to optimize the neural network to prevent inconsistency with the optimization direction of the expected performance parameters. Therefore, to the above loss function, this study added an additional constraint term, as shown in Fig. 7. The PyTorch version was used to calculate the average sound absorption coefficient corresponding to the geometric structure parameters predicted by the DNN model. In addition, the expected average sound absorption coefficient was determined, and its mean square error was calculated as follows:

$$L_\alpha = \frac{1}{M} \sum_{i=1}^M (\bar{\alpha}_i - \bar{\alpha}_i'), \quad (19)$$

The total loss function was expressed by:

$$L = L_s + L_\alpha, \quad (20)$$

4. Prediction results and verification

After training the DNN model using the sample database constructed in this study, we randomly generated two curves with different variations, both within the prediction frequency range of 1–1000 Hz, as shown in Fig. 8. The curve characteristics of case 1 are continuously

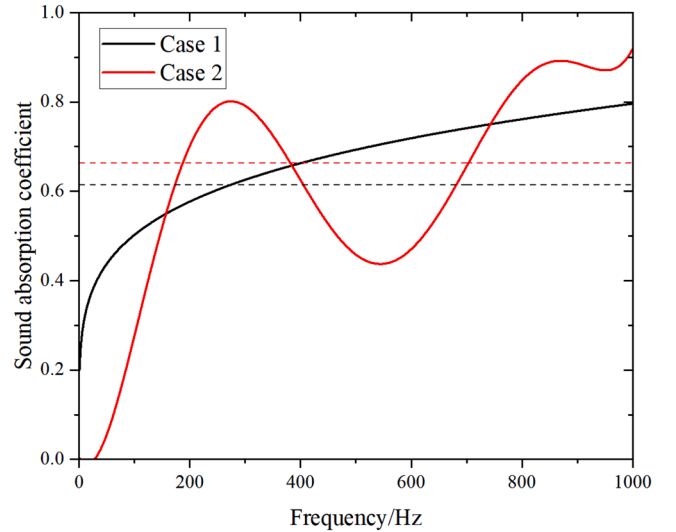


Fig. 8. The two randomly selected absorption coefficient curves.

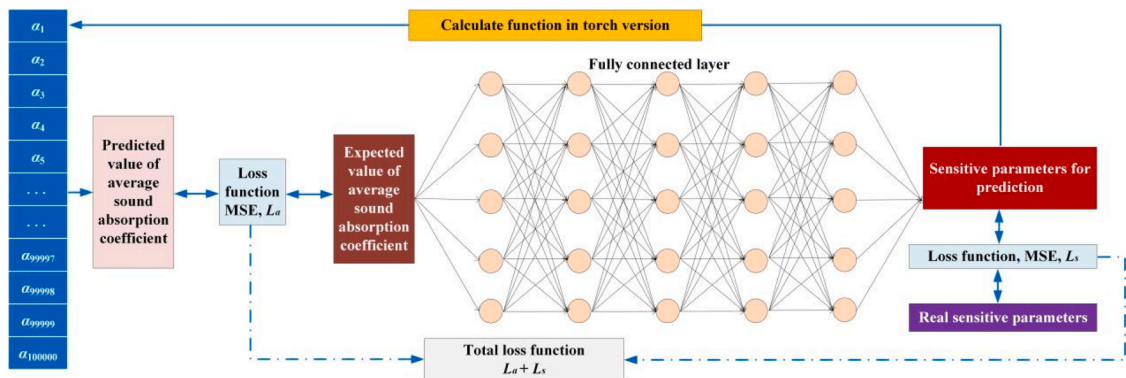


Fig. 7. The flow diagram of the DNN with the MSE loss.

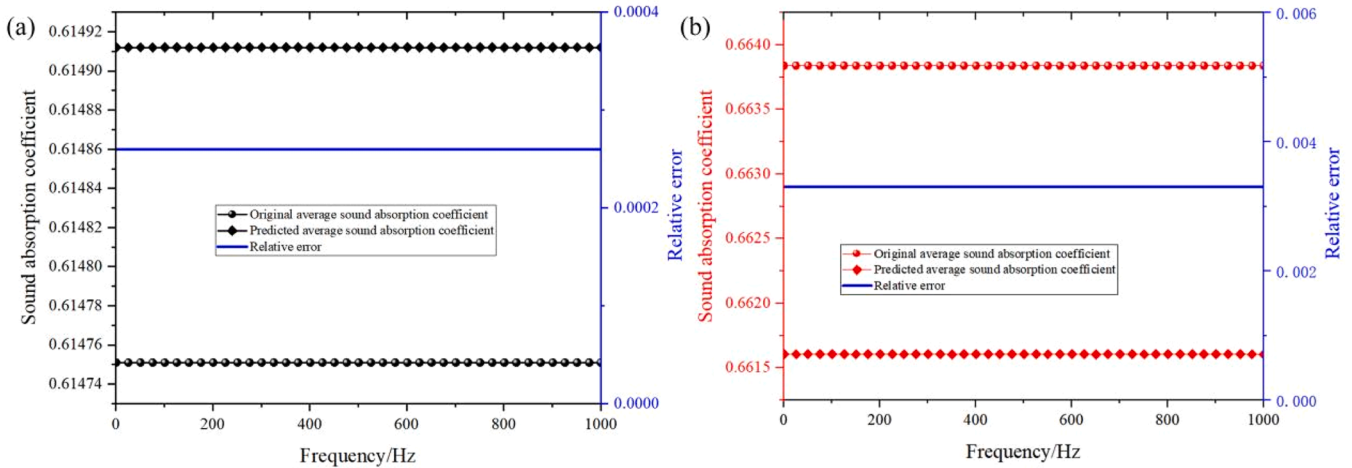


Fig. 9. Relative error between original and predicted average sound absorption coefficients.

Table 5

The prediction results of the 20 sensitivity parameters of the average value of the predicted sound absorption curve.

Prediction results for Case 1			
d_1 (mm)	10.7398927211761	d_2 (mm)	9.19723930954933
d_3 (mm)	9.03188413381577	d_4 (mm)	10.3210442066193
d_5 (mm)	9.02042958140373	d_6 (mm)	9.03090962767601
d_7 (mm)	10.1851529777050	d_8 (mm)	9.20814347267151
d_9 (mm)	9.10563176870346	d_{10} (mm)	10.3094107210636
m_2 (mm)	353.101807236671	m_3 (mm)	354.557039141655
m_5 (mm)	371.865028738976	m_6 (mm)	382.696526646614
m_8 (mm)	404.766833186150	m_9 (mm)	400.110272169113
ρ (kg/m ³)	1268.43154430389	η	0.758901304006577
E (Pa)	8.49030417203903E7(1+ η i)	ν	0.476384522914887
Prediction results for Case 2			
d_1 (mm)	10.4116438329220	d_2 (mm)	9.59269911050797
d_3 (mm)	9.48593208193779	d_4 (mm)	10.2778358161449
d_5 (mm)	9.56153133511543	d_6 (mm)	9.52644741535187
d_7 (mm)	10.2548802793026	d_8 (mm)	9.57734596729279
d_9 (mm)	9.54049196839333	d_{10} (mm)	10.3517957925797
m_2 (mm)	422.048478722572	m_3 (mm)	415.016191601753
m_5 (mm)	421.283624768257	m_6 (mm)	428.282207846642
m_8 (mm)	437.154577970505	m_9 (mm)	433.320407271385
ρ (kg/m ³)	1254.90951538086	η	0.700865012407303
E (Pa)	7.96552193164825E7(1+ η i)	ν	0.472864060401917

increasing from 1 Hz, while the curve characteristics of case 2 are both increasing and decreasing within the predicted frequency range, and neither of these curves belong to the curves in the database in Fig. 6. The distributions of the two absorption curves did not exceed the envelope of the sample database presented in Fig. 6. The sound absorption coefficients corresponding to the two curves were 0.614751 and 0.663840, as indicated by the dashed lines in Fig. 8. The two average absorption coefficients were input into the neural network of Section 3, and the above method was used for prediction to obtain two predicted absorption coefficient curves and the corresponding 20 sensitive parameters. The average values of the predicted sound absorption coefficient curves were 0.614912 and 0.661605, with relative errors of 0.026 % and 0.33 %, respectively. Their distributions and errors are shown in Fig. 9a and b. Obviously, the reliability of the prediction results was very high. The 20 sensitive parameters corresponding to the predicted average sound absorption coefficient, achieve very high accuracy to ten decimal places, and are all listed in Table 5.

5. Conclusion

This study introduces a DNN model with an improved loss function

for predicting an average absorption coefficient curve. The overall loss function of the DNN model includes the original loss function aided with the MSE-based loss function. The MSE value is calculated between the expected average absorption coefficient and its predicted value, targeting the 100,000 randomly generated absorption coefficient curves in the theoretical acoustic performance calculation model of underwater coating; this represents a sample database constructed using 20 sensitive parameters, including 16 geometric parameters and four material parameters. Considering that inputting a set of performance parameters in the development framework might correspond to multiple sets of 20 sensitive parameters, using the original loss function could prevent DNNs from obtaining optimal results. Therefore, the on-demand design presented in this paper adds additional constraints to the DNN model to determine whether it can achieve the best unique results. By performing the prediction tests for two randomly selected absorption coefficient curves, the effectiveness of the proposed method is verified, demonstrating very small errors of 0.026 % and 0.33 %. The work presented in this article could provide a valuable reference for accelerating the performance design of underwater acoustic materials.

CRedit authorship contribution statement

Nansha Gao: Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mou Wang:** Software, Resources, Formal analysis. **Xiao Liang:** Methodology. **Guang Pan:** Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

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Data availability

Data will be made available on request.

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